Maths Homework Template

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Short Math Guide: `http://tug.ctan.org/info/short-math-guide`.

Short Math Guide (Chinese version): `https://github.com/hoganbin/short-math-guide`.

1. ¶ Inline equations for simple math: $\boldsymbol{p} = \hbar \boldsymbol{k}$, $\varepsilon = \hbar \boldsymbol{\omega}$, $\phi = \dfrac{1+\sqrt{5}}{2}$.

   ¶ Centered equations for more involved or important equations:

$$\boldsymbol{a} \cdot \boldsymbol{b} = \sum_{i=1}^{n} a_i b_i \tag{1}$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \mathrm{e}^{a(x)} = a'(x) \mathrm{e}^{a(x)} \tag{2}$$

$$\frac{\|\mathsf{A}\boldsymbol{x}\|_\infty}{\|\boldsymbol{x}\|_\infty} = \frac{\max\limits_i \left| \sum\limits_{j=1}^{n} \mathsf{A}_{ij} \mathsf{x}_{j1} \right|}{\max\limits_i |\mathsf{x}_{i1}|} \tag{3}$$

2. ¶ An example of a matrix:

$$\mathsf{M} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \\ a_1^2 & a_2^2 & \cdots & a_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \cdots & a_n^{n-1} \end{bmatrix}, \quad \det(\mathsf{M}) = \prod_{1 \leqslant j \leqslant i \leqslant n} (a_i - a_j) \tag{4}$$

   ¶ An example of accents:

$$\hat{h} \quad \check{c} \quad \tilde{t} \quad \acute{a} \quad \grave{g} \quad \dot{d} \quad \ddot{d} \quad \dddot{d} \quad \ddddot{d} \quad \breve{b} \quad \bar{b} \quad \vec{v} \tag{5}$$

   ¶ An example of math fonts:

$$\begin{matrix} \mathrm{ABCD} & \mathit{ABCD} & \boldsymbol{ABCD} & \mathsf{ABCD} \\ \mathcal{ABCD} & \mathscr{ABCD} & \mathfrak{ABCD} & \mathbb{ABCD} \end{matrix} \tag{6}$$

   ¶ An example of roots:

$$r = \sqrt[3]{\frac{3}{1 + \sqrt{\frac{3}{1+\sqrt{3}}}}} \tag{7}$$

   ¶ An example of Dirac notation:

$$\begin{matrix} \langle \phi | \psi \rangle & \left\langle \frac{1}{2}\lambda \middle| \psi \right\rangle & \left| \phi_{\hat{Q}=0} \right\rangle \\ \langle A \rangle & \left\langle \frac{1}{2}\lambda \right\rangle & \langle \Psi | A | \Psi \rangle \end{matrix} \tag{8}$$

3.

$$\iiint \nabla \cdot \boldsymbol{A}\,\mathrm{d}V = \oiint \boldsymbol{A}\,\mathrm{d}\boldsymbol{S}$$

4.

$$F_n(z) = \begin{cases} \dfrac{1}{\sqrt{1+in^2\mathrm{e}^{Az}}}, & z < 0 \\ \dfrac{1}{\sqrt{1+n^2\mathrm{e}^{Az}}}, & z \geqslant 0 \end{cases}$$

5.

$$u(x,t) = \sum_{n=0}^{\infty} X_n(x)Y_n(y) = \frac{8a^2}{\pi^3}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^3}\left(1 - \frac{\cosh\dfrac{2k+1}{a}\pi y}{\cosh\dfrac{2k+1}{2a}\pi b}\right)\sin\frac{2k+1}{a}\pi x$$

6.

$$\begin{aligned} \mathscr{L}\left\{\frac{1-\cos\omega t}{t}\right\} &= \int_p^{\infty}\left(\frac{1}{p} - \frac{1}{2(p+\mathrm{i}\omega)} - \frac{1}{2(p-\mathrm{i}\omega)}\right)\mathrm{d}t \\ &= \ln p - \frac{1}{2}\ln(p+\mathrm{i}\omega) - \frac{1}{2}\ln(p-\mathrm{i}\omega)\Big|_p^{\infty} \\ &= \frac{1}{2}\ln\frac{p^2+\omega^2}{p^2} \end{aligned}$$

7.

$$\begin{aligned} [q,p]_{Q,P} &= \frac{\partial q}{\partial Q}\cdot\frac{\partial q}{\partial P} - \frac{\partial q}{\partial P}\cdot\frac{\partial q}{\partial Q} \\ &= 1 + \frac{P^2\mathrm{e}^{2Q}}{1-P^2\mathrm{e}^{2Q}} - \frac{P^2\mathrm{e}^{2Q}}{1-P^2\mathrm{e}^{2Q}} \\ &= 1 \end{aligned}$$

8.

$$\frac{4\pi R_\odot^2\cdot\sigma T_\odot^4}{4\pi d^2} = 1352.83\,\mathrm{W/m^2}$$

$$T_\odot = 5763.43\,\mathrm{K}$$

$$\lambda_\odot = \frac{b}{T_\odot} = 5027.86\,\text{Å}$$

9.

$$\begin{aligned} I(x) &= \int_{-\frac{b}{2}}^{\frac{b}{2}}\mathrm{d}I(x) \\ &= 4\alpha\frac{I_0}{b}\int_{-\frac{b}{2}}^{\frac{b}{2}}\cos^2\left(\frac{\pi d}{\lambda D}\left(x+\frac{D}{R}\xi\right)\right)\mathrm{d}\xi \\ &= 4\alpha I_0\left(1+\frac{1}{u}\sin u\cos\left(\frac{2\pi d}{\lambda D}x\right)\right), \quad u = \frac{\pi d b}{\lambda R} \end{aligned}$$